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23 APRIL 1951
SUPERSEDING
JAN-I-631
10 August 1948

MILITARY SPECIFICATION

INSULATION, ELECTRICAL, SYNTHETIC-RESIN COMPOSITION, NONRIGID

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.—This specification covers only the basic characteristics required by the synthetic insulating material, as such, and the methods suited to their satisfactory determination.

1.2 Classification. — Electrical insulation covered by this specification shall be of the following types, forms, grades, classes, and categories, as specified (see 6.1 and 6.2):

Type A—Polyethylene.

Type B—Cellulose acetate butyrate.

Type C—Cellulose acetate.

Type D—Ethyl cellulose.

Type E—Vinylidene chloride copolymer.

Type F—Polyvinyl chloride and its copolymers.

Forms.—Electrical insulation shall be furnished in the following forms, as specified (see 6.2):

Form F—Film, in rolls (types B, C and D only).

Form S—Sheet, flat or in rolls.

Form T—Tape.

Subform T_f—Film tape (types B, C, and D only).

Subform T_s—Sheet tape.

Form U—Tubing (see 6.6).

Subform U_a—American-wire-gage size (AWG size) (types A, E and F only).

Subform U_b—Cable-end-sealing size (types F only).

Grades.—Type F electrical insulation shall be furnished in the following grades, as specified (see 6.2):

Grade a—General purpose.

Grade b—Low temperature (form U only).

Grade c—High temperature (form U only).

Classes.—Electrical insulation shall be furnished in the following classes, as specified (see 6.2):

Class I—Fungus resistance required.

Class II—Fungus resistance not required.

Categories.—Form U electrical insulation shall be furnished in the following categories, as specified (see 6.2):

Category 1—Flame resistance required.

Category 2—Flame resistance not required.

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(Copies of the Navy Shipment Marking Handbook should be obtained from the sources given for obtaining specifications for the Navy.)

MILITARY STANDARD

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military standards should be obtained from the sources given for obtaining specifications.)

2.3 Drawings.—The following drawing, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SHIPS DRAWING

9–S–5357–L—Sealing, Cable Ends.

(Copies of Bureau of Ships drawings may be obtained *only* upon application to the Bureau of Ships, Navy Department, Washington 25, D. C. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Qualification. — Electrical insulation furnished under this specification shall be a product which has been tested and has successfully passed the qualification tests specified in section 4. (See 6.3.)

3.2 Material.—The material shall be of a suitable synthetic-resin composition properly formulated with plasticizers, pigment, stabilizers, or other ingredients necessary to conform to this specification.

3.2.1 *Changes in material.*—If the manufacturer desires to change his process or the composition of his material, after qualification approval has been granted (see 6.3) he shall submit to the qualifying agency (Bureau of Ships) a statement to that effect, describing the proposed change, together with evidence to substantiate his claim that such change is suitable. Similar notification and substantiating evidence shall be submitted at any later time if a change again becomes necessary or desirable. At the discretion of the qualifying agency, test specimens may be required to prove the suitability of the proposed changes.

3.2.2 *Uniformity.*—All material in a lot of the same type and brand of any one manufacturer shall be uniform in texture, color, cross section, finish, and in the properties specified in tables I, II, or III, as applicable.

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2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 **Specifications.**—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATIONS

NN–B–601 —Boxes, Wood-Cleated-Plywood, for Domestic Shipment.

NN–B–621 —Boxes, Wood, Nailed and Lock-Corner.

NN–B–631 —Boxes, Wood, Wire-bound, for Domestic Shipment.

VV–O–401 —Oil; Insulating (for Transformers, Switches, and Circuit-Breakers).

LLL–B–631—Boxes, Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN–P–108—Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

JAN–P–127—Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water Resistant.

JAN–P–139—Packaging and Packing for Overseas Shipment — Plywood, Container Grade.

JAN–P–140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

U. S. ARMY SPECIFICATIONS

94–40645 —Marking; Exterior, Domestic and Export Shipment, by Contractors.

100–2 —Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.

U. S. AIR FORCE SPECIFICATION

40985 —Marking of Interior Packages (for Communications Equipment).

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal, Military and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Federal and Military specifications may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 **Other publications.** — The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.

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TABLE I.—*Property values of synthetic-resin electrical insulation—forms F and T (0.002 inch and less in thickness).*

Property to be tested	Test paragraph	Conditioning procedure	Unit of value	Nominal thickness	Value required for each type of material		
					B	C	D
				Inches			
Dielectric strength[1]	4.6.2	C–96/25/0	Volts per mil (min.)	0.0005	2,700	2,700	3,500
				.001	2,700	2,700	3,500
				.0015	2,350	2,350	3,300
				.002	2,150	2,150	3,000
		C–96/25/96			[2]	[3]	[4]
Dissipation factor (form F only)							
at 1 kilocycle	4.6.3	C–96/25/0	Maximum	----------	0.02	0.035	0.015
		C–96/25/96			.03	.045	.03
at 1 megacycle	----------	C–96/25/0	Maximum	----------	0.045	.05	.028
		C–96/25/96			.055	.07	.045
Dielectric constant (form F only)							
at 1 kilocycle	4.6.3.	C–96/25/0	Maximum	----------	4.0	4.5	4.0
		C–96/25/96			5.0	8.0	5.0
at 1 megacycle	----------	C–96/25/0	Maximum	----------	4.0	4.3	4.0
		C–96/25/96			5.0	7.0	5.0
Volume resistivity (form F only)	4.6.4.1	C–96/25/0	Ohm-centimeters (min.)	----------	10^{13}	10^{13}	10^{14}
		C–96/25/96			10^{12}	10^{12}	10^{12}
Tensile strength	4.6.5.1	C–96/25/50	p.s.i. (min.)	----------	5,000	6,000	3,500
Elongation	4.6.5.1	C–96/25/50	Percent (min.)	----------	25	10	25
		C–96/25/50	Percent(max.)	----------	100	50	100
Softening temperature (penetration)	4.6.7	C–96/25/50	Degrees centigrade (min.)	----------	120	160	110
Lengthwise shrinkage	4.6.8	E–2/100	Percent (max.)	----------	3.0	3.0	2.0

[1] For thickness not shown, the dielectric strength shall be that given for the nearest thickness shown.
[2] 80 percent of dry value obtained on test.
[3] 65 percent of dry value obtained on test.
[4] 55 percent of dry value obtained on test.

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TABLE II.—*Property values of synthetic-resin electrical insulation—forms S and T (greater than 0.002 inch in thickness).*

Property to be tested	Test paragraph	Conditioning procedure (see 4.4)	Unit of value	Nominal thickness	Value required for each type of material					
					A	B	C	D	E	F Grade a
				Inches						
Dielectric strength[1]	4.6.2	C–96/25/0	Volts per mil (min.)	0.003	3,100	1,900	1,900	2,600	1,200	1,400
				.004	2,750	1,700	1,700	2,300	1,120	1,400
				.005	2,400	1,550	1,550	2,150	1,070	1,400
				.008	1,800	1,300	1,250	1,800	960	1,130
				.010	1,600	1,200	1,150	1,550	920	1,000
				.015	1,300	1,000	950	1,200	940	820
				.020	1,100	900	840	1,050	790	700
				.025	960	820	760	920	740	630
				.030	870	760	700	830	700	570
				.040	740	640	620	700	600	480
				.050	660	550	550	610	530	430
				.060	600	490	490	540	450	400
		C–96/25/96			[2]	[2]	[3]	[4]	[5]	[6]
Dissipation factor (form S only)										
At 1 kilocycle	4.6.3	C–96/25/0	Maximum		0.0006	0.02	0.035	0.015	0.08	0.14
		C–96/25/96			.0008	.03	.045	.03	.08	.16
At 1 megacycle		C–96/25/0	Maximum		.0006	.045	.05	.028	------	--------
		C–96/25/96			.0008	.055	.07	.045	------	--------
Dielectric constant (form S only)										
At 1 kilocycle	4.6.3	C–96/25/0	Maximum		2.5	4.0	4.5	4.0	5.0	7.0
		C–96/25/96			2.7	5.0	8.0	5.0	6.0	8.0
A 1 megacycle		C–96/25/0	Maximum		2.5	4.0	4.3	4.0	------	--------
		C–96/25/96			2.7	5.0	7.0	5.0	------	--------
Volume resistivity (for S only)	4.6.4.1	C–96/25/0	Ohm-centimeters (min.)		10^{17}	10^{13}	10^{13}	10^{14}	10^{13}	10^{11}
		C–96/25/96			10^{16}	10^{12}	10^{12}	10^{12}	10^{12}	10^{10}
Tensile strength	4.6.5.1	C–96/25/50	p.s.i. (min.)		1,400	5,000	6,000	3,500	3,500	1,800
Elongation	4.6.5.1	C–96/25/50	Percent (min.)		350	25	10	25	15	200
		C–96/25/50	Percent (max.)		-------	100	50	100	100	450
		E–400/100	Percent change (max.)		-------	------	------	------	------	35
Percent return	4.6.5.2	C–96/25/50	Percent (min.)		-------	------	------	------	------	80
Softening temperature (penetration)	4.6.7	C–96/25/50	Degrees centigrade (min.)		65	120	160	110	120	75
Lengthwise shrinkage	4.6.8	E–2/100	Percent (max.)		2.0	3.0	3.0	2.0	5.0	10.0

[1] For thickness not shown, the dielectric strength shall be that given for the nearest thickness shown.
[2] 85 percent of dry value obtained on test.
[3] 75 percent of dry value obtained on test.
[4] 55 percent of dry value obtained on test.
[5] 95 percent of dry value obtained on test.
[6] 90 percent of dry value obtained on test.

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TABLE III.—*Property values of synthetic-resin electrical insulation form U.*

Property to be tested	Test paragraph	Conditioning procedure	Unit of value	Nominal dimensions	Value required for each type of material				
					A	E	F Grade a	F Grade b	F Grade c
				Inches Wall thickness					
Dielectric strength[1]	4.6.2	C–96/25/0	Volts per mil (min.)	0.015	1,250	1,820	800	750	800
				.020	1,100	790	700	650	700
				.025	960	740	630	580	630
				.030	870	700	570	520	570
		C–96/25/96		.040	740	600	480	430	480
				.050	660	530	430	380	430
				.060	600	450	400	350	400
				------	[2]	[3]	[4]	[5]	[6]
Dissipation factor									
at 1 kilocycle	4.6.3	C–96/25/0	Maximum	------	0.0006	0.08	0.14	0.14	0.14
		C–96/25/96			.0008	.08	.16	.16	.16
at 1 megacycle	------	C–96/25/0	Maximum	------	.0006	------	------	------	------
		C–96/25/96			.0008	------	------	------	------
Dielectric constant									
at 1 kilocycle	4.6.3	C–96/25/0	Maximum	------	2.5	5.0	7.0	7.0	7.0
		C–96/25/96			2.7	6.0	8.0	8.0	8.0
at 1 megacycle	------	C–96/25/0	Maximum	------	2.5	------	------	------	------
		C–96/25/96			2.7	------	------	------	------
Volume resistivity	4.6.4.2	Dt–24/25	Ohm-centimeters (min.)	------	------	------	10^{10}	10^{10}	10^{11}
Tensile strength	4.6.5.1	C–96/25/50	p.s.i. (min.)	------	1,400	3,500	1,800	1,450	1,800
		C–96/25/50	Percent (min.)	------	350	15	200	200	200
		C–96/25/50	Percent (max.)	------	------	100	450	450	450
Elongation	4.6.5.1	E–400/100	Percent change (max.)	------	------	------	35	35	------
		E–400/130	Percent change (max.)	------	------	------	------	------	35
Percent return	4.6.5.2	C–96/25/50	Percent (min.)	------	------	------	70	70	65

[1] For thickness not shown, the dielectric strength shall be that given for the nearest thickness shown.
[2] 85 percent of dry value obtained on test.
[3] 95 percent of dry value obtained on test.
[4] 90 per cent of dry value obtained on test.
[5] 85 percent of dry value obtained on test.
[6] 85 percent of dry value obtained on test.

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TABLE III.—*Property values of synthetic-resin electrical insulation form U.*—Continued

Property to be tested	Test paragraph	Conditioning procedure	Unit of value	Nominal dimensions	Value required for each type of material				
					A	E	F Grade a	F Grade b	F Grade c
				Inside diameter (min.)					
				0.155	--------	--------	96	96	96
				.185	--------	--------	80	80	80
				.245	--------	--------	59	59	59
				.275	--------	--------	53	53	53
				.315	--------	--------	46	46	46
				.355	--------	--------	41	41	41
				.395	--------	--------	58	58	58
				.425	--------	--------	54	54	54
				.445	--------	--------	50	50	50
				.515	--------	--------	44	44	44
				.565	--------	--------	54	54	54
				.625	--------	--------	49	49	49
Bursting strength (subform U_b only) increasing pressure	4.6.6	C–96/25/50	p.s.i. (min.)	.675	--------	--------	45	45	45
				.750	--------	--------	41	41	41
				.810	--------	--------	47	47	47
				.880	--------	--------	43	43	43
				.950	--------	--------	40	40	40
				1.020	--------	--------	38	38	38
				1.090	--------	--------	35	35	35
				1.150	--------	--------	40	40	40
				1.205	--------	--------	38	38	38
				1.260	--------	--------	36	36	36
				1.345	--------	--------	34	34	34
				1.455	--------	--------	32	32	32
				1.565	--------	--------	29	29	29
Continuous pressure	4.6.6	C–96/25/50	p.s.i. (min.)	--------	--------	--------	[7]	[7]	[7]
Softening temperature (penetration)	4.6.7	C–96/25/50	Degrees centigrade (min.)	--------	65	120	75	70	80
Lengthwise shrinkage	4.6.8	E–2/100	Percent (max.)	--------	5.0	5.0	10.0	10.0	--------
		E–130		--------	--------	--------	--------	--------	10.0
Cold brittleness	4.6.12	--------	Degrees centigrade	--------	−60	−24	−30	−46	−10

[7] One-half of above value.

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3.2.3 *Surface defects.*—Materials shall be free from blisters, bubbles, wrinkles, and cracks, and shall be reasonably free from other small defects such as scratches and dents.

3.2.4 *Colors.*—Materials shall be furnished in the colors specified (see 6.2).

3.3 Dimensions and tolerances.

3.3.1 Film and film tape (form F and subform T_f).

3.3.1.1 Film is defined as material in thicknesses of 0.002 inch or less, in widths greater than 3 inches. Material 3 inches or less in width, slit from film, is defined as film tape (subform T_f).

3.3.1.2 *Thicknesses.* — Unless otherwise specified, the standard thicknesses of form F material, and of subform T_f material cut from film, shall be as shown in table IV as specified (see 6.2).

TABLE IV.—*Standard thicknesses of film and film tape (form F and subform T_f)*

Thickness	Tolerance
Inch	*Inch*
0.0005	± 0.00025
.001	± .00025
.0015	± .00025
.002	+ .0003 — .00025

3.3.2 *Sheet and sheet tape (form S and subform T_s).*

3.3.2.1 Sheet is defined as material in thicknesses greater than 0.002 inch and in widths greater than 3 inches. Material 3 inches or less in width, slit from sheet, is defined as sheet tape (subform T_s). Thin sheets shall be supplied in rolls, if specified (see 6.2).

3.3.2.2 *Thicknesses.* — Unless otherwise specified, the nominal thicknesses and tolerances on thicknesses of form S material, and of subform T_s material cut from sheets, shall be as follows, as specified (see 6.2):

Nominal thickness	Tolerance (plus or minus)	Nominal thickness	Tolerance (plus or minus)
Inch	Inch	Inch	Inch
0.003	0.0003	0.020	0.0020
.004	.0004	.025	.0025
.005	.0005	.030	.0030
.008	.0008	.040	.0040
.010	.0010	.050	.0050
.015	.0015	.060	.0060

At least 90 percent of the area of the sheet shall be within this variation, and at no point shall the thickness as measured vary from the nominal by a value greater than 12.5 percent.

3.3.3 *Tape.*—Tape shall be furnished in any specified thickness, in rolls of 3 inches wide or less, wound over a suitable cylindrical core for subsequent application by hand or machine taping (see 6.2). The thicknesses and thickness tolerances for extruded tapes shall be as specified in 3.3.1.2, 3.3.2.2, as applicable and as specified (see 6.2).

3.3.3.1 *General.*—Tape shall be uniform, even, and smooth in appearance, with a surface entirely free from holes. The edges shall be straight, true, and unbroken. It shall be tightly wound in rolls, with a suitable separator between layers, when necessary, to prevent the material from adhering to itself when unwound from the roll. When unwound from the roll, the tape shall remain reasonably flat with no tendency to curl transversely. The tape, when correctly applied, shall adhere snugly without wrinkling, curling, or buckling.

3.3.3.2 *For application by machine.*—When the material is required for machine-taping, the width, tolerance on the width, and length of material on the roll, the over-all diameter of the roll, and the mandrel diameter of the

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cylindrical core shall be as specified, (see 6.2).

3.3.3.3 *For application by hand.*—When the material is required for hand-taping, it shall, unless otherwise specified, be put up in rolls approximately 4½ inches in diameter, wound over a suitable 1- or 1½-inch core convenient for hand application. Unless otherwise specified, tape for hand application shall be furnished in the standard widths shown in table V, as specified. (See 6.2).

TABLE V.—*Standard widths of tape for application by hand (form T)*

Width	Tolerance (plus or minus)
Inches	*Inch*
0.50	0.02
.75	.02
1.00	.02
1.25	.04
1.50	.04
2.00	.04
2.50	.04
3.00	.04

3.3.4 *Tubing (form U).*

3.3.4.1 Unless otherwise specified, tubing shall be furnished in lengths of 25 feet (see 6.2). If continuous lengths greater than 25 feet are specified, they shall be furnished on reels in sizes 11 to 20 AWG. All other sizes shall be furnished in coils (see 6.2).

3.3.4.2 Inside diameters and wall thicknesses.—Unless otherwise specified, tubing shall be furnished in the standard inside diameters and wall thicknesses shown in table VI, as specified (see 6.2).

3.4 Quantitative requirements (property values).—When specimens are subjected to the tests specified in table I, II, or III, as applicable, the values obtained from any set of specimens for any specific property after any one conditioning procedure specified in the table shall be averaged, and this average value shall be not less than the minimum or more than the maximum value (as applicable) specified in table I, II, or III, as applicable.

3.5 Qualitative requirements.

3.5.1 *Effect of heat on materials under tension (forms F, S, and T).*—The material shall not crack or split in layers when tested as specified in 4.6.9.

3.5.2 *Flexibility after aging.*

3.5.2.1 *Oven aging.*—None of the samples of material shall exhibit cracks due to hardening when tested as specified in 4.6.10.2. In the testing of form U material, cracking at the creased edges of the bend shall be disregarded.

3.5.2.2 *Ultra-violet-light aging (type F, grade b only).*—Type F, grade b material shall not exhibit cracks due to hardening when tested as specified in 4.6.10.3. In the testing of form U material, cracking at the creased edges of the bend shall be disregarded.

3.5.3 *Cold bend (forms F, S and T only).*—The material shall not crack or break when tested as specified in 4.6.11.

3.5.4 *Flame resistance form U only (category 1) (see 1.2).* — The average duration of burning shall not exceed 15 seconds, and the paper indicator shall show no evidence of being affected when tested as specified in 4.6.13.

3.5.5 *Oil resistance-flexibility (not applicable to types A and D).*

3.5.5.1 *Forms F, S and T.*—The material shall not increase in dimension (width or thickness) by more than 15 percent when tested as specified in 4.6.14. It shall not break or crack when creased on itself.

3.5.5.2 *Form U.*—At the completion of the test specified in 4.6.14, oil shall not have penetrated into the tubing; and the tubing shall not have developed cracks, or increased in diameter at any point by more than 10 percent.

3.5.6 *Corrosive effect.*—Copper shall not be corroded or otherwise adversely affected by contact with the material. (See 4.6.15.)

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TABLE VI.—*Inside diameters and wall thicknesses of tubing, (form U).*

Size No.	AWG size (subform U_a)					
	Inside diameter (inches)			Wall thickness (inches)		
	Nominal	Minimum	Maximum	Minimum	Maximum	Nominal
20	0.034	0.032	0.039	0.013	0.019	0.016
19	.038	.036	.044	.013	.019	.016
18	.042	.040	.049	.013	.019	.016
17	.047	.045	.054	.013	.019	.016
16	.053	.051	.061	.013	.019	.016
15	.059	.057	.067	.013	.019	.016
14	.066	.064	.072	.013	.019	.016
13	.076	.072	.080	.013	.019	.016
12	.085	.081	.089	.013	.019	.016
11	.095	.091	.101	.013	.019	.016
10	.106	.102	.112	.013	.019	.016
9	.118	.114	.124	.017	.023	.020
8	.133	.129	.141	.017	.023	.020
7	.148	.144	.158	.017	.023	.020
6	.166	.162	.178	.017	.023	.020
5	.186	.182	.198	.017	.023	.020
4	.208	.204	.224	.017	.023	.020
3	.234	.229	.249	.017	.023	.020
2	.263	.258	.275	.017	.023	.020
1	.294	.289	.311	.017	.023	.020
0	.330	.325	.347	.017	.023	.020
5/16"	----------	.313	.334	.022	.028	.025
3/8"	----------	.375	.399	.022	.028	.025
7/16"	----------	.438	.462	.022	.028	.025
1/2"	----------	.500	.524	.022	.028	.025
5/8"	----------	.625	.655	.025	.035	.030
3/4"	----------	.750	.786	.030	.040	.035
7/8"	----------	.875	.911	.030	.040	.035
1"	----------	1.000	1.036	.030	.040	.035
1 1/4"	----------	1.250	1.290	.035	.045	.040
1 1/2"	----------	1.500	1.550	.039	.051	.045
1 3/4"	----------	1.750	1.812	.047	.063	.055
2"	----------	2.000	2.070	.050	.070	.060
Cable-end-sealing size[1] (subform U_b)—type F only.						
	0.162	0.155	0.170	0.018	0.023	0.020
	.190	.185	.200	.018	.023	.020
	.250	.245	.260	.018	.023	.020
	.280	.275	.290	.018	.023	.020
	.320	.315	.330	.018	.023	.020
	.360	.355	.375	.018	.023	.020
	.400	.395	.420	.028	.033	.030
	.430	.425	.450	.028	.033	.030
	.450	.445	.470	.028	.033	.030
	.520	.515	.540	.028	.033	.030
	.570	.565	.590	.037	.043	.040
	.630	.625	.650	.037	.043	.040
	.680	.675	.700	.037	.043	.040
	.760	.750	.785	.037	.043	.040
	.820	.810	.845	.046	.054	.050
	.890	.880	.925	.046	.054	.050
	.960	.950	1.000	.046	.054	.050
	1.030	1.020	1.070	.046	.054	.050
	1.100	1.090	1.150	.046	.054	.050
	1.165	1.150	1.200	.055	.065	.060
	1.225	1.205	1.255	.055	.065	.060
	1.285	1.260	1.310	.055	.065	.060
	1.365	1.345	1.400	.055	.065	.060
	1.470	1.455	1.510	.055	.065	.060
	1.585	1.565	1.615	.055	.065	.060

[1] Included only for information of Navy Department personnel applying tubing in Navy applications covered by Drawing 9–S–5357–L.

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3.5.7 *Fungus resistance (class I only) (see 6.4).*—The material shall be tested for fungus resistance as specified in 4.6.16. The material shall not support fungus growth; this shall be due to absence of nutritive substances in the material and not to presence of a fungistatic agent. The material shall be considered fungus resistant if:

a. At least 2 of 3 specimens are rated 0 to 1 when tested as received (4.6.16.5(a)) and
b. At least 2 of 3 specimens are rated 0 to 1 when tested after being conditioned for 6 hours at 85° C. (4.6.16.5(b)).

3.6 Chemical analysis.—The results of the chemical analysis of material offered under the contract or order shall be approximately the same as that of samples submitted for qualification approval. (See 4.6.17).

3.7 Workmanship.—The material shall be manufactured and processed in a careful and workmanlike manner, in accordance with good practice.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Qualification tests at a Government laboratory.—Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.3.

4.2 Classification of tests.—The methods of sampling, inspection, and tests shall be classified as follows:

Qualification tests. (See 4.3.)
Inspection tests. (See 4.4.)

4.3 Qualification test.

4.3.1 *Samples.*—Separate samples for each type, form, and subform[1] of material for which approval is requested shall be furnished in the sizes and quantities shown in table VII. (See 4.3.2.2.)

4.3.2 *Tests.*

4.3.2.1 All samples shall be subjected to the tests shown in table VIII, except as specified in 4.3.2.2.

4.3.2.2 *Tape cut from film or sheet.*—Qualification testing will not be required on tape which is formed by slitting film or sheet when qualification for such film or sheet has already been obtained. Complete qualification testing will be required on tape formed directly by extrusion or on tape slit from film or sheet for which qualification approval has not been obtained (see 4.3.1).

4.3.3 *Failures.*—Failure of any of the samples in any one of the tests for each type, form, and subform submitted may be cause for refusal to grant qualification approval for that particular type, form, or subform.

4.4 Inspection and tests.—The insulation shall be subjected to inspection by authorized Government inspectors. When tests are conducted at the manufacturer's plant, they shall be performed by the manufacturer under the supervision of the Government inspector.

4.4.1 *Sampling.*

4.4.1.1 *Lot.*—A lot shall consist of all material of the same type, form, and thickness (forms F, S, and T) or inside and outside diameters (form U), offered for delivery at the same time.

4.4.1.2 *Sampling for lot acceptance inspection.*—A random sample of rolls, sheets, coils, or reels of insulation shall be selected from each lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with MIL–STD–105[2]:

[1] Because of the large range of sizes in forms T and U, complete qualification approval for these forms is not necessarily required. As an alternative, at the option of the manufacturer, qualification approval will be required on the individual subforms of forms T and U for which it is desired (see 1.2, table VII, and 4.3.2.2).

[2] At the option of the Government inspector, the equivalent double or sequential sampling plan from MIL–STD–105 may be used.

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TABLE VII.—*Sizes and quantities of material for qualification tests*

Form	Sizes			Width or inside diameter (minimum)	Amount of each size
	Sub-form	Size number	Thickness		
			Inch	*Inches*	
F	----------	----------	0.0005	----------	10 square yards of each thickness
		----------	.001	----------	
		----------	.002	----------	
S	----------	----------	0.005	----------	10 square yards of each thickness
		----------	.010	----------	
		----------	.020	----------	
T*	T_f*	----------	0.0005	1	1,000 feet of each thickness
		----------	.001	1	
		----------	.002	1	
	T_s*	----------	.005	1	
		----------	.010	1	
		----------	.020	1	
U	U_a Type F only	18	----------	0.040	200 feet
		13	----------	.072	200 feet
		6	----------	.162	200 feet
	U_b Type F only	----------	----------	1.020	100 feet
		----------			

* One square yard of sheet or film material in each thickness will also be required when tape is obtained by slitting (see 4.3.2.2). The material shall be furnished in any convenient width over 3½ inches.

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TABLE VIII.—*Qualification tests*

Test	Requirement paragraph	Test paragraph	Minimum number of tests for each condition (see 4.5 to 4.5.2, inclusive).
General inspection	3.2 to 3.3.4.2, inclusive, and 3.7	4.6.1	10
Electrical properties:			
Dielectric strength	3.4	4.6.2	10
Dissipation factor (forms F, S and U only)	3.4	4.6.3	3
Dielectric constant (forms F, S and U only)	3.4	4.6.3	3
Volume resistivity:			
Forms F and S only	3.4	4.6.4.1	3
Type F, form U only	3.4	4.6.4.2	3
Mechanical properties:			
Tensile strength	3.4	4.6.5.1	10
Elongation	3.4	4.6.5.1	10
Percent return (type F only)	3.4	4.6.5.2	3
Bursting strength (type F, form U only)	3.4	4.6.6	3
Thermal properties:			
Softening temperature (penetration)	3.4	4.6.7	5
Lengthwise shrinkage	3.4	4.6.8	3
Effect of heat on materials under tension (forms F, S and T only)	3.5.1	4.6.9	2
Flexibility:			
After oven aging	3.5.2.1	4.6.10.2	3
After ultra-violet-light aging (type F, grade b only)	3.5.2.2	4.6.10.3	4
Cold bend (forms F, S, and T only)	3.5.3	4.6.11	2
Cold brittleness (form U only)	3.4	4.6.12	as specified in 4.6.12.3
Flame resistance (form U only)	3.5.4	4.6.13	5
Miscellaneous:			
Oil resistance—flexibility (not applicable to types A and D)	3.5.5	4.6.14	3
Corrosive effect	3.5.6	4.6.15	2
Fungus resistance	3.5.7	4.6.16	6
Chemical analysis	3.6	4.6.17	1

4.4.1.3 *Sampling for lot-acceptance tests.*—The Government inspector shall select from each lot the number of samples shown in table X, for tests as specified in 4.4.3.

4.4.2 *Lot acceptance inspection (at place of manufacture).*—The Government inspector shall subject the samples (see 4.4.1.2) to a thorough inspection to make certain that the material and workmanship conform to 3.2.2 to 3.2.4, inclusive, and 3.7, and that the dimensions are as specified in 3.3 to 3.3.4.2. If the number of defective samples exceeds the acceptance number for the appropriate sampling plan in table IX, the lot represented by the samples shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming material. The Government inspector shall again select and examine samples from such rejected lots to verify compliance with the specification requirements.

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TABLE IX.—*Sampling for lot acceptance inspection AQL (approx.) = 1.2 to 2.2 percent defective*

Number of rolls, sheets, coils or reels in lot	Number of rolls, sheets, coils or reels to be sampled	Acceptance number (defectives)	Rejection number (defectives)
1 to 10	All	--	--
11 to 25	10	0	1
26 to 50	15	0	1
51 to 100	20	1	2
101 to 200	30	2	3
201 to 300	40	2	3
301 to 500	55	3	4
501 and over	75	3	4

TABLE X.—*Number of samples for lot-acceptance tests*

Number of rolls, sheets, coils or reels in lot[1]	Number of separate rolls, sheets, coils, or reels to be sampled[2]
1 to 3	All
Over 3	3

[1] No samples shall be taken from lots of tape consisting of 100 rolls or less or lots of tubing consisting of 1,500 feet or less (see 4.4.3.2).

[2] The minimum quantity of each sample shall be as follows:

Forms F and S—9 square feet
Form T—1 roll, or a minimum of 40 feet
Form U—25 feet

4.4.3 *Lot acceptance tests (at a Government laboratory).*

4.4.3.1 Lot acceptance tests (except as specified in 4.4.3.2), shall be performed at the Government laboratory specified by the bureau or service concerned. The tests shall consist of those shown in table XI, all samples selected in accordance with 4.4.1.3 shall be subjected to the group 1 tests, and two samples selected in accordance with 4.4.1.3 shall be subjected to the group 2 tests. If one or more of the samples fail one or more of the tests, the entire lot shall be rejected.

4.4.3.2 *Lot acceptance tests for small orders.*—Lots of tape consisting of 100 rolls or less, or lots of tubing consisting of 1,500 feet or less, shall be tested for acceptance at the place of manufacture. Only those tests will be required by the Government inspector which are considered necessary to establish essential equivalence with the approved quality. If one or more of the samples fail one or more of the tests, the entire lot will be rejected.

4.4.4 *Action in case of lot rejection.*—A rejected lot may be resubmitted for lot acceptance inspection and lot acceptance tests only after the manufacturer has reinspected and retested the entire lot for the deficiency noted and has removed all nonconforming material.

4.5 **Conditioning.**—Materials shall be conditioned before being tested, as specified in table I, II, or III, as applicable.

4.5.1 *Equipment.*—Conditioning chambers and baths shall be such as to maintain the specified temperature plus or minus 2° C. and the specified relative humidity plus or minus 2 percent.

4.5.2 *Designation.* — Conditioning procedures shall be designated as follows:

a. A capital letter indicating the general condition of the specimen; i.e., humidity, immersion, and temperature conditioning.

b. A number indicating in hours the duration of the conditioning.

c. A number indicating in degrees centigrade the conditioning temperature.

d. A number indicating relative humidity whenever relative humidity is controlled. (Relative humidity obtained over calcium chloride shall be taken as zero.)

The numbers shall be separated from each other by slant marks, and from the capital letter by a dash.

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TABLE XI.—*Lot acceptance tests*

Test	Requirement paragraph	Test paragraph	Conditioning prior to test (see 4.5).	Minimum number of tests per sample
Group 1—all samples				
Dielectric strength	3.4	4.6.2	C–96/25/0	3
			C–96/25/96	3
Tensile strength and elongation	3.4	4.6.5.1	C–96/25/50	2
Percent return (type F, form U only)	3.4	4.6.5.2	C–96/25/50	1
Bursting strength (type F, Form U_b only), continuous pressure	3.4	4.6.6	C–96/25/50	1
Lengthwise shrinkage	----------	4.6.8	E–2/100 or E–2/130 as specified in tables I, II, or III.	1
Group 2—two samples				
Dissipation factor and dielectric constant (forms F, S, and U), at 1 kilocycle only[1]	3.4	4.6.3	C–96/25/96	3
Volume resistivity (forms F and S)[1]	3.4	4.6.4	C–96/25/96	3
Softening temperature (penetration)	3.4	4.6.7	C–96/25/50	5
Flexibility after oven aging	3.5.2.1	4.6.10.2	E–200/100	2
Cold bend (forms F, S, and T only)	3.5.3	4.6.11	----------	2
Cold brittleness (form U only)	3.4	4.6.12	----------	as specified in 4.6.12.3
Chemical analysis	3.6	4.6.17	----------	see footnote [2]

[1] Not required on types E and F.

[2] One composite sample.

4.5.3 *Nomenclature.*—The following letters shall be used to indicate the respective general conditioning:

Condition C—The specimens shall be conditioned for the time and at the temperature and relative humidity specified in table I, II, or III, as applicable.

Examples:

C–96/25/0—96 hours, 25° C., 0 percent relative humidity (desiccation over calcium chloride).

C–96/25/50—96 hours, 25° C., 50 percent relative humidity.

C–96/25/96—96 hours, 25° C., 96 percent relative humidity.

Condition D_t—The specimens shall be immersed in 1 percent salt water for the time and at the temperature specified.

Example:

D_t–24/25—24 hours, 25° C., in 1 percent salt water.

Condition E—The specimens shall be conditioned for the time and at the temperature specified.

Examples:

E–400/100—400 hours, 100° C.
E–200/100—200 hours, 100° C.
E–2/100—2 hours, 100° C.
E–400/130—400 hours, 130° C.

4.6 **Tests.**—The methods and procedures shall be as specified in 4.6.1 to 4.6.17, inclusive. Unless otherwise specified, all tests shall be conducted under room conditions (25° C., 50 percent relative humidity). In the case of materials believed to be oriented, mechanical tests shall be conducted on samples cut both lengthwise and crosswise, for the properties affected by the orientation.

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4.6.1 *General inspection.*

4.6.1.1 The insulation shall be inspected to verify that the materials, dimensions, and workmanship are in strict conformance with this specification, (see 3.2 to 3.3.4.2, inclusive and 3.7.).

4.6.1.2 *Methods for measurement of dimensions.*

4.6.1.2.1 *Thickness (forms F, S and T).*—The thickness shall be determined by the use of a dial-type micrometer. The lower or fixed anvil shall consist of a flat steel plate 2 inches in diameter; the upper or movable member shall consist of a foot or button, having a flat steel face measuring ¼ inch in diameter, and shall be arranged so as to move with its face parallel to the face of the anvil at all times. The movable member shall be of such weight, and shall be attached to the operating rod of the micrometer in such a manner, that 3 ounces dead weight shall be applied to all thicknesses of samples being measured. The thickness shall be determined as the average of at least 10 readings taken at representative points on the material. If the thickness so determined falls outside the specified tolerances, 10 additional readings shall be taken from another section of the material and averaged with the previous readings. The maximum, minimum, and average values shall be recorded.

4.6.1.2.2 *Width (form T).*—The width shall be determined by the use of a standard steel scale graduated to read within 0.01 inch. The tape shall be flat on a smooth surface at the time of measurement. At least 10 measurements shall be made at representative points on the material, and the maximum, minimum, and average values shall be recorded.

4.6.1.2.3 *Diameter and wall thickness (form U).*—A plug, slightly smaller in diameter than the inside diameter of the tubing, shall be inserted into a small section of tubing to spread it and form an approximate ellipse without stretching. Using a toolmaker's microscope, the inside and outside dimensions of the tubing shall be measured at both axes. The wall thickness shall be calculated from these measurements and the tubing diameter shall be the sum of the major and minor axes divided by two.

NOTE: If a toolmaker's microscope is not available at the manufacturer's plant for inspection tests, plug gages may be used for measuring the inside diameter of tubing. In case of dispute, however, measurements made with a toolmaker's microscope shall govern.

4.6.2 *Dielectric strength.*

4.6.2.1 *Test equipment.*—The test equipment shall consist of a highvoltage transformer rated at not less than 2 kilovolt-amperes and the necessary auxiliary equipment for applying, controlling, and measuring the test voltage. The power supply shall consist of an alternating-current source having as nearly a true sine wave as possible at a frequency not exceeding 100 cycles per second.

4.6.2.2 *Voltage control.*—The high-tension voltage taken from the secondary of the testing transformer shall be capable of being raised gradually from any point and in no case more than 500 volts at a step. The control may be made by generator field regulation with an induction regulator, or with a variable ratio autotransformer. Any method of controlling the voltage is satisfactory which does not distort the wave more than 10 percent from a sinusoidal shape.

4.6.2.3 *Voltage measurement.*—The voltage may be measured by any approved method which gives root-mean-square values, preferably by means of a voltmeter connected to a tertiary coil in the testing transformer, or to a separate step-down instrument potential transformer. A volmeter on the low tension side of the transformer is satisfactory if the ratio of the transformation does not change under any test condition. An electrostatic

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voltmeter properly calibrated in the high-tension circuit is also satisfactory. A spark gap may be used to check the readings at very high potentials.

4.6.2.4 *Test electrodes.*

4.6.2.4.1 *Forms F and S.*—The test electrodes for forms F and S materials shall consist of metal cylinders 2 inches in diameter and not less than ½ inch in length, with edges rounded to a radius of ¼ inch. The total weight on the sample shall be approximately 1 pound.

4.6.2.4.2 *Form T.*—The test electrodes for form T material shall consist of bars 4 inches long and ¼ inch wide, rounded at the ends, and with all edges in contact with the sample rounded to a radius of $\frac{1}{32}$ inch. The total weight on the sample shall be approximately ½ pound.

4.6.2.4.3 *Form U.*—The test electrodes for form U material shall consist of a solid conductor which fits the inside of the tubing closely without stretching the tubing, and tightly wrapped 1-inch strip of metal foil on the outside. In the case of flexible tubing of 0.315-inch inside diameter and larger, the tubing may be cut along its axis, flattened, and tested with the electrodes specified in 4.6.2.4.2.

4.6.2.5 *Specimens.* — The test specimens shall be 3½ inches square for forms F and S material; 6 inches long by the width of the material for form T material; and 6 inches long for form U material. Tests shall be conducted at 25° C. immediately after removal of the material from conditions C–96/25/0 and C–96/25/96.

4.6.2.6 *Procedure.*

4.6.2.6.1 Tests shall be made with the specimen immersed in temperature-controlled transformer oil conforming to Specification VV–0–401. This oil shall be changed frequently to prevent excessive deterioration. The test electrodes shall be kept clean and polished. They shall be self-aligning and shall be placed exactly opposite one another, with the specimen in a horizontal plane between them.

4.6.2.6.2 Starting at zero, the voltage shall be increased uniformly to break-down, at a rate of 500 volts per second. The average dielectric break-down of the material shall be expressed in volts. The dielectric strength shall be computed by dividing the average dielectric break-down by the average thickness and shall be reported in volts per mil.

4.6.3 *Dissipation factor and dielectric constant*[3] (*Forms F, S and U only*).

4.6.3.1 *Test equipment.* — The test equipment shall be a suitable bridge or resonant-circuit.

4.6.3.1.1 *Bridge method.*—The bridge or bridges used shall be suitable for the measurement of dissipation factor and dielectric constant at the specified frequencies. Five bridge arrangements which have been found satisfactory for the purpose intended are the "Conjugate Schering Bridge," the "High Voltage Schering Bridge," the "Parallel Resistance Bridge," the "Series Resistance Bridge," and the "Transformer Bridge." It is not intended, however, that the use of other types of bridges of equal precision and convenience be excluded.

4.6.3.1.2 *Resonant-circuit method.* — The resonant circuit used shall be of a type suitable for the measurement of dissipation factor and dielectric constant at 1 megacycle. Three resonant-circuit arrangements which have been found satisfactory for the purpose intended are the "Susceptance Variation," the "Resistance Variation," and "Resonant Rise." It is not intended however, that the use of

[3] For a complete discussion of the theory, apparatus, electrodes, procedure and calculations, refer to A.S.T.M.D 150–46 T, Tentative Methods of Test for Power Factor and Dielectric Constant of Electrical Insulating Materials.

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other types of resonant circuit arrangements of equal precision and convenience be excluded.

4.6.3.1.3 *Sensitivity.* — The method used shall have an accuracy of determination of dielectric constant of plus or minus 5 percent, and an accuracy of determination of dissipation factor of plus or minus 5 percent or 0.0002, whichever is greater.

4.6.3.2 *Specimens.*—Where possible, specimens should be in the form of flat sheets. Larger sizes (1 inch or greater) of tubing may be slit along the axis of the tubing and flattened to obtain a sheet specimen. Suitable coaxial electrodes may be used on small tubing.

4.6.3.3 *Electrodes.*—One electrode shall be of metal foil, 3 inches in diameter, on one surface of the specimen. The electrode on the other surface shall be of metal foil 2⅝ inches in diameter. When possible a concentric, metal-foil guard electrode ⅛ inch wide and 3 inches in outside diameter shall be used. The electrodes shall be applied to the specimen by using a very thin film of petrolatum thoroughly rolling the electrodes so applied to exclude air.

4.6.4 *Volume resistivity.*

4.6.4.1 *Forms F and S (only).*

4.6.4.1.1 *Apparatus.*—The apparatus shall consist of a source of direct-current potential of 100 to 200 volts, a galvanometer with suitable shunts having a minimum sensitivity of 0.0005 microampere per millimeter at 1 meter-scale distance, a calibrating resistance, reversing switches, and keys. Any other apparatus may be used provided the same accuracy is obtained and the same voltage applied.

4.6.4.1.2 *Specimens.*—The size of the test specimens shall be 4 by 4 inches by thickness.

4.6.4.1.3 *Electrodes.*—The electrodes shall be as specified in 4.6.3.3.

4.6.4.1.4 *Procedure.*—The electrodes shall be applied to the specimen, and the volume resistance between the guarded and unguarded electrodes shall be measured by means of a test set which applies approximately 180 volts direct current to the specimen. The reading shall be taken after voltage has been applied for 1 minute.

4.6.4.1.5 *Calculation.*—Volume resistivity, r, in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where:

R = volume resistance in megohms.
A = area of guarded electrode in square centimeters.
t = average thickness of specimen in centimeters.

4.6.4.2 *Form U (type F only).*

4.6.4.2.1 *Apparatus.*—The apparatus shall consist of that specified in 4.6.4.1.1; in addition, there shall be a container for water large enough to accommodate a specimen of tubing bent into a U-shape, and a support for the two branches of the U-shaped tubing.

4.6.4.2.2 *Specimens.*—The test specimens shall be cut sufficiently long so that a satisfactory reading may be obtained on the galvanometer. The minimum length shall be 2 feet.

4.6.4.2.3 *Procedure.* — The specimen shall be shaped without kinking into a U-form and immersed in water, at 25° ±2° C., containing 1 percent sodium chloride. Not less than 1 foot of the specimen shall be submerged. To prevent leakage of electric current, the ends of the specimen shall protrude 6 inches above, and approximately perpendicular to the surface of the water. The specimen shall be filled immediately with the same salt water, and, as a precaution against air bubbles and discontinuity of the water phase, a fine wire shall be inserted throughout the

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length of the specimen. The salt water in the tubing shall serve as an inner electrode, and its surface shall be above the surface of the salt water in the container. The salt water in the container shall serve as an outer electrode. The inner and outer electrodes shall be connected to the circuit. The specimen shall remain in the water for 24 hours. At the end of this time the reading shall be taken.

4.6.4.2.4 *Calculation.*—Volume resistivity, r, in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RL \times 10^6}{0.3663 \times \log_{base} 10 \frac{D}{d}}$$

Where:

R = volume resistance in megohms.
L = length of specimen in centimeters.
D = outside diameter.
d = inside diameter.

4.6.5 *Tensile strength, elongation, and percent return.*

4.6.5.1 *Tensile strength and elongation.*

4.6.5.1.1 *Apparatus.*—Tensile strength and elongation shall be determined on an approved pendulum-type or constant rate of elongation testing machine. The machine shall preferably be power-driven.

4.6.5.1.2 *Specimens.* — Ten specimens of each size of material shall be tested after being subjected to conditions C–96/25/50, E–400/100, and E–400/130 when required in tables I, II and III. Specimens cut from form F, and from forms S and T measuring 0.010 inch and less in thickness, shall be 1 inch wide. Specimens cut from forms S and T measuring more than 0.010 inch in thickness shall be ½ inch, dumbbell, die-cut specimens, except that tape less than 1½ inches wide shall be tested in full section. Form U tubing of 0.315-inch inside diameter and larger shall be tested by using ½-inch, dumbbell, die-cut specimens. Form U tubing in smaller size inside diameters shall be tested in full tubing section.

4.6.5.1.3 *Procedure.*

4.6.5.1.3.1 Bench marks 10 inches apart shall be used for materials having ultimate elongations of 25 percent or less. Bench marks 2 inches apart shall be used for materials having ultimate elongations greater than 25 percent. On aged specimens, bench marks shall be applied prior to aging.

4.6.5.1.3.2 The test specimens shall be placed in the testing machine and stretched until they break. The rate of separation of the jaws of the machine shall be 12±½ inch per minute. A scale shall be used to measure the elongation at the instant of rupture. The tensile strength shall be reported in pounds per square inch.

4.6.5.2 *Percent return (type F only).*—The tests shall be conducted on specimens similar to those specified in 4.6.5.1.2 which have been subjected to condition C–96/25/50.

4.6.5.2.1 *Procedure.*

4.6.5.2.1.1 The specimens shall be stretched an amount equal to 50 percent of the elongation value after condition C–96/25/50, obtained when tested as specified in 4.6.5.1.3.2. The stretch shall be maintained for 100 seconds, and the material shall be allowed to recover for 60 seconds. Three samples shall be tested. The distance between bench marks and the rate of elongation shall be as specified in 4.6.5.1.3.1.

4.6.5.2.1.2 *Calculation.* — The percent return shall be expressed as the distance returned, multiplied by 100 and divided by the distance stretched.

4.6.6 *Bursting strength (type F, form U only).*

4.6.6.1 *Apparatus.* — Bursting strength shall be measured on 9-inch lengths of tubing clamped over a source of compressed air at one end and plugged at the other end in such a manner that the clamps are 6 inches apart

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and the tubing cannot expand in a lengthwise direction.

4.6.6.2 *Specimens.*—Three samples of each size of tubing shall be subjected to each of the tests specified in 4.6.6.3.

4.6.6.3 *Procedure.*

4.6.6.3.1 *Increasing pressure.*—The pressure shall be increased in steps of 1 pound at the rate of 5 pounds per minute, until failure.

4.6.6.3.2 *Continuous pressure.*—The pressure shall be increased in steps of 1 pound at the rate of 5 pounds per minute, until the desired continuous pressure shown in table III is attained. This pressure shall be maintained for 24 hours.

4.6.7 *Softening temperature (penetration).*

4.6.7.1 *Apparatus.* — The apparatus shall consist of an oven capable of raising the temperature of a stainless steel plate ⅛ by 1¼ by 4 inches, at the rate of 1° C. every 2 minutes; a $\frac{1}{16}$-inch magnetized steel rod, recessed at one end to hold a $\frac{1}{16}$-inch ball bearing, in contact with and equipped to exert a pressure of 1,000 grams vertically down against the stainless steel plate, lying in a horizontal plane; an electrical circuit with a 110-volt source of electricity; a 110-volt glow lamp; and a suitable means of measuring the temperature of the steel plate as near the point of test application as practicable. The use of a thermocouple to measure this temperature is suggested. Equipment, as shown on figure 1, is recommended. The light C-clamp, containing the steel rod, counterbalance, and weight, shall be mounted on a bearing capable of giving the unit the necessary freedom of clockwise motion. With no load on the lower leg of the C-clamp, the counterbalance shall be adjusted to neutralize the pressure of the ball bearing against the steel plate.

4.6.7.2 *Specimens.*—Five 1-inch specimens shall be cut from the sample and prepared for test. For tubing, this may be accomplished by opening one side along the longitudinal axis.

4.6.7.3 *Procedure.*—With no load on the rod, each specimen shall be inserted between the steel ball and steel plate; tubing shall be inserted with its outside surface facing the plate. The electrical circuit shall be connected in such a way that when the steel ball comes in contact with the plate, the lamp contained outside the oven will light. The specimen in the oven at room temperature shall then be put under a compression load of 1,000 grams. The temperature of the steel plate shall be uniformly raised at the rate of 1° C. every 2 minutes until failure of the specimen, which is indicated by the lighting of the lamp. The temperature at failure shall then be recorded as the softening temperature (penetration), to determine compliance with applicable requirements of table I, II, or III.

4.6.8 *Lengthwise shrinkage.* ....

4.6.8.1 *Specimens.* — Three specimens of each size of material, in strips ½ inch wide by 6 inches long with 5 inches between bench marks, shall be tested. Strips cut from forms F and S material shall be used. Tape and tubing shall be tested in full section.

4.6.8.2 *Procedure.*

4.6.8.2.1 *Forms F, S, and T.*—The specimens shall be supported in an oven by a smooth-surfaced, thin metal plate covered with talc, and subjected to condition E–2/100. The specimens shall then be removed and cooled to room temperature (25° C.). The length of the material shall be measured before and after heating, and the percentage of shrinkage calculated from these values.

4.6.8.2.2 *Form U.*—The specimens shall be supported by insertion of a solid conductor having a diameter slightly less than the inside diameter of the tubing, and well coated with talc to prevent bonding. The specimens

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so supported shall be held horizontally in an oven and subjected to condition E–2/100 or E–2/130, as specified in table III. The specimens shall then be removed and cooled to room temperature (25° C.). The length of the material shall be measured before and after heating and the percent shrinkage calculated from these values.

4.6.9 *Effect of heat on material under tension (forms F, S, and T only).*—Brass tubing approximately 1 inch in diameter and 3¾ inches long shall be wrapped with two half-lapped windings of material approximately 1 inch wide under a tension of 500 pounds per square inch, and the end secured. The specimens shall be subjected to condition E–400/100 and examined at intervals for cracking or splitting of layers.

4.6.10 *Flexibility after aging.*

4.6.10.1 *Specimens.*—A sufficient number of 1-inch-width strips of forms F and S material and of form T tape wider than 1 inch shall be used so that samples can be discarded after bending. Tape 1 inch wide and less, and form U tubing without conductor inside, shall be tested in full section.

4.6.10.2 *Oven aging.*—The specimens shall be subjected to condition E–200/100 in a forced-draft oven, cooled to room temperature (25° C.), and bent as follows:

Forms F, S, and T.—The material shall be bent 180 degrees around a mandrel of a diameter equal to its own thickness.

Form U.—The tubing, in full section without conductor inside, shall be bent 180 degrees around a ⅛-inch-diameter mandrel.

4.6.10.3 *Ultra-violet-light aging (type F, grade b only).*

4.6.10.3.1 *Apparatus.*—The specimens shall be secured to a light-colored, corrosion-resistant metal disk, approximately 17 inches in diameter and 0.1 inch in thickness, mounted on a phonograph turntable operating at 33 revolutions per minute. The disk shall be centered under an S–1 bulb which has been in use not less than 50 hours and not more than 550 hours, and which is mounted in a General Electric sun lamp, model BM12, or an equivalent lamp, equipped with a reflector approximately 15 inches in diameter at the lower rim. The lamp and turntable shall not be enclosed. The plane of the specimens shall be 6 inches from the bottom of the bulb.

4.6.10.3.2 *Mounting of specimens.* — The specimens shall be supported about 3/16 inch above the surface of the disk by a wire screen. Specimens shall be 4 inches long by not more than 1½ inches in width or diameter. The specimens shall be mounted radially with their inside end 3 inches from the center of the disk.

4.6.10.3.3 *Procedure.* — After exposure to the ultraviolet radiation for 120 hours, the specimens shall be removed and bent as specified in 4.6.10.2.

4.6.11 *Cold bend (applicable only to forms F, S and T).*

4.6.11.1 *Specimens.*—Strips of film, sheet and tape, 1 inch wide shall be used. Tape less than 1 inch wide shall be tested in full width.

4.6.11.2 *Procedure.* — Specimens shall be conditioned for 1 hour in air at –23° C. All specimens shall be bent while at the required temperature, as specified in 4.6.10.2.

4.6.12 *Cold brittleness (Form U only).*

4.6.12.1 *Specimens.*—For tubing 0.148-inch inside diameter and larger test specimens ¼ inch in width and 1½ inches in length shall be cut from the tubing. This shall be accomplished by cutting a ¼ inch strip along a longitudinal axis of the sample. For tubing smaller than 0.148-inch inside diameter, test specimens in full section 1½ inches in length shall be cut from the tubing.

4.6.12.2 *Apparatus.*[4]—The apparatus may be either of the single specimen or multiple specimen type. The specimen clamp, designed

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to hold the specimen as a cantilever beam with half the length unsupported and the striking edge shall be capable of motion relative to each other at a constant linear speed of 6.5 ± 0.5 feet per second at and during impact. The distance between the center line of the striking edge and the clamp shall be 0.31 ± 0.01 inch at impact. The striking edge shall have a radius of 0.062 ± 0.005 inch. The striking arm and specimen clamp shall have a clearance of 0.25 ± 0.01 inch at and after impact. These dimensional requirements for the recommended apparatus are shown on figure 2. The apparatus shall also include:

a. A thermometer of suitable temperature range graduated in 1° C. divisions.
b. A liquid coolant which will not affect the material under test.
c. Dry ice for cooling the coolant-dry ice mixture.
d. A tank for holding the coolant dry-ice mixture and the test mechanism.
e. A stirrer for efficient and rapid mixing of the coolant.

[4] One possible design of apparatus is shown in A.S.T.M. D746–44T, Tentative Method of Test for Brittle Temperature of Plastics and Elastomers.

4.6.12.3 *Procedure.*—For qualification test, 18 specimens shall be tested at each temperature starting at the estimated brittle temperature. A range of temperatures shall be covered such that both the 0 percent breaks (18 nonbreaks) and the 100 percent (18 breaks) are included. Two degrees C. temperature-intervals shall be used and the above range shall be completely covered. The machine shall be operated so that the impact blow is delivered as specified in 4.6.12.2. For lot acceptance tests, 18 specimens shall be selected from each type and each size submitted. Where there is more than one sample of any one type or size, the 18 specimens shall be equally distributed between these samples. The specimens shall be immersed in the coolant at the applicable temperature in table III plus or minus .5° C. for 2.5 ± .5 minutes, and the blow delivered as specified in 4.6.12.2.

4.6.12.4 *Evaluation.*—Failure is defined as the separation of the specimen into two or more separate pieces. The occurrence of fissures or cracks does not constitute brittle-type failure although their presence shall be reported. For qualification tests the 50% brittle temperature shall be calculated. The following equation shall be used:

$$m = X + \frac{2S_1}{100} - 1$$

Where:

m = 50 percent brittle temperature, °C.
X = Highest temperature at which 100 percent of specimens break (use proper algebraic sign, minus in most cases).
S_1 = Sum of percentages of breaks at each temperature (from a temperature corresponding to 0 percent breaks to and including X).

For lot acceptance tests not more than 9 of the 18 specimens shall break.

4.6.13 *Flame resistance (form U only category 1) (see 1.2).*

4.6.13.1 *Test apparatus.*

4.6.13.1.1 *Sheet-metal enclosure.*—A three-wall sheet-metal enclosure, 12 inches wide by 14 inches long by 29 inches high, shall be used. The walls facing each other shall support two parallel horizontal steel rods; these rods of ¼ inch diameter shall be separated 16 to 21 inches as required, with the lower rod closer to the back wall, and shall be located so that they define a plane which shall be set at an angle of 70 degrees to the horizontal.

4.6.13.1.2 *Bare steel wire.*—A length of bare steel wire, approximately 0.029 inch in diameter for all sizes of specimens, shall be attached at one of its ends to the center of the upper rod. The other end of the wire shall pass over the center of the lower rod and shall have a 1-pound weight attached to maintain the wire in a straight line across the two rods.

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4.6.13.1.3 *Tirrill burner.*—A Tirrill burner having a nominal bore of 3/8 inch and a length of approximately 4 inches above the primary inlets, with an attached pilot light, shall be used.

4.6.13.1.4 *Gas supply.*—A supply of ordinary illuminating gas at normal pressure shall be available.

4.6.13.1.5 *Timer.*—A timer measuring seconds shall be used.

4.6.13.1.6 *Flame indicators.* — Strips of gummed paper shall be used as flame indicators.

4.6.13.2 *Test specimens.*—For inside diameters of 7/8 inch and less, five test specimens 17 inches long shall be cut from the sample. For inside diameters of more than 7/8 inch, five test specimens 22 inches long shall be cut from the sample.

4.6.13.3 *Procedure.*

4.6.13.3.1 *For round tubing.*

4.6.13.3.1.1 *Preparation.*—The test shall be made in a room or cabinet free from drafts of air. The specimen shall be drawn onto the wire. The covered wire shall be fastened into position, with its length adjusted so that the weight hangs free, maintaining tension and supporting the specimen in a straight line at an angle of 70 degrees with the horizontal. One end of the specimen shall be near the lower rod, and the other end shall be drawn to the upper rod and clamped so that the specimen is closed, in order to prevent a draft through the tube. The paper indicator shall be applied to the upper end of the specimen, in such a position that, for tubing 7/8 inch and less in inside diameter, the lower edge is 10 inches away from the point at which the inner cone of the test flame is to be applied. For tubing more than 7/8 inch in inside diameter, the lower edge of the paper indicator shall be 15 inches away from the point at which the inner cone of the test flame is to be applied. The indicator shall be wrapped and pasted around the specimen.

4.6.13.3.1.2 *Application of flame.* — The height of the flame with the burner in a vertical position shall be adjusted to 5 inches with an inner cone 1½ inches high. The burner, with only the pilot lighted, on a base at an angle of 25 degrees with the horizontal, shall be placed in such a position under the specimen that the axis of the burner makes an angle of 95 degrees with the specimen and there is a distance of 1½ inches between the tip of the stem and the surface of the specimen, along the axis of the burner stem. The valve controlling the gas supply shall be opened, and the flame applied to the specimen for 15 seconds.

4.6.13.3.2 *For flat tubing.*

4.6.13.3.2.1 *Preparation.*—The wire shall support the specimen by one of the two small curvatures found in flat tubing. The flats of the tubing shall be directed vertically downward. The distance of 1½ inches from burner to specimen shall be between the tip of the stem and the surface of the narrow side of the flat specimen.

4.6.13.3.2.2 *Application of flame.* — The valve controlling the gas supply to the burner shall be opened, and the flame shall be applied to the narrow side of the specimen for a period of 15 seconds. The duration of burning shall be determined immediately after removal of the flame.

4.6.13.3.3 *Calculation.* — The length of specimen burned shall be determined by subtracting the length of the unburned portion from the original length of the test specimen. The unburned portion is the distance from the lower edge of the indicator paper to the first obvious defect caused by the flame.

4.6.14 *Oil resistance—flexibility (not applicable to types A and D).*

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4.6.14.1 *Specimens.*—Three specimens shall be tested.

4.6.14.2 *Apparatus.*—Tests on material in forms F, S, T, and U shall be made in a closed container approximately 6 inches deep, filled with No. 3 high-swelling oil[5] to a line 2 inches below the cover. Strips of material in forms F, S, and T shall be hung in straight lengths with one end protruding through the cover and the lower end below the surface of the oil. Specimens of material in form U shall be bent in the shape of a U around a diameter of approximately 4 inches, with the ends protruding through the cover and the bottom of the U located 3 inches below the surface of the oil. The specimen then has a section immersed in oil, a section in oil vapor, and a section above the cover subjected to oven temperature only.

[5] High-swelling oil is characterized by Saybolt universal viscosity of 155 ± 5 seconds at 100°F., aniline point 157.1° ± 18°F., and flash point 330° ± 5°F., as described in A.S.T.M. D471–46 T, Tentative Methods of Test for Changes in Properties of Rubber and Rubber-Like Materials in Liquids.

4.6.14.3 *Procedure.*—The entire equipment shall be subjected to condition E–200/50. The specimens shall then be removed and tested to determine compliance with 3.5.5.

4.6.15 *Corrosive effect.*—Forms F, S, and T, and form U tubing of 0.315-inch inside diameter and larger, shall be tested by butt-wrapping a ½-inch strip full length and return on two 6-inch piece of 0.162-inch diameter (AWG No. 6) copper wire. Tubing of less than 0.315-inch inside diameter shall be tested by inserting the proper size wire into a 6-inch length of the tubing. A 30-inch length of 0.006-inch diameter, bare copper wire in a bifilar winding shall then be wrapped over the prepared specimens. The ends of the wire shall be welded to heavy leads of 0.040-inch diameter (AWG No. 18) copper wire, and the sample placed in a test tube with the leads extending through a cork. A small amount of distilled water shall be placed in the test tube to maintain a high relative humidity, and the cork shall be coated to produce a vaportight seal. The entire equipment shall be placed in an oven maintained at 70° C. The resistance of the 0.006-inch diameter copper wire shall be measured when the temperature of the specimen reaches 70° C. and at 24-hour intervals for 720 hours. At the end of this period, specimens shall be examined for visual evidence of corrosion on the 0.006-inch diameter wire. Percentage change in resistance versus time shall be plotted to show corrosion characteristics of the material.

4.6.16 *Fungus resistance (class I—only) (see 1.2).*

4.6.16.1 *Test fungi.* — The following test fungi (see 6.5) shall be used:

	American Type Culture Collection Number
Aspergillus niger	9642
Aspergillus flavus	9643
Penicillium luteum	9644
Trichoderma T–1	9645

Cultures of these fungi shall be maintained separately on an appropriate medium, such as potato-dextrose agar. The stock cultures may be kept for not more than 4 months in a refrigerator, at approximately 3° to 10° C. Subcultures incubated at 28° to 30° C., for 7 to 20 days, shall be used in preparing the spore suspension.

4.6.16.2 *Spore suspension.* — A separate sterile 10-ml. portion, of distilled water or of a solution containing per liter 0.05 gm. of a non-toxic wetting agent such as dioctyl sodium sulfosuccinate, shall be poured into one subculture of each of the four fungi and agitated to dislodge the spores. The resulting separate spore suspensions shall be mixed to provide a composite spore suspension for use in the test. The spore suspension may be prepared freshly upon the day of the test, or may be held in the refrigerator at 3° to 10° C. for not more than 7 days.

4.6.16.3 *Controls.* — The viability of the

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inoculum shall be checked with each composite spore suspension, upon each day when it is used for inoculation. Three agar slants, of an appropriate medium such as potato-dextrose agar, shall serve as controls. A drop of the composite spore suspension shall be deposited with aseptic precautions at the top of each slant. The controls shall then be incubated with the test specimens. The spore suspension shall be considered satisfactory provided all three controls are covered with fungus growth at the end of the incubation period of 21 days.

4.6.16.4 *Specimens.* — Six specimens shall be used. Specimens of film, sheet, or tape shall consist of portions approximately 1½ inches square. Specimens of large-size tubing shall be 1½-inch squares cut from the tubing; specimens from small-diameter tubing shall be 1½-inch lengths cut from the tubing.

4.6.16.5 *Conditioning.* — Conditioning of specimens shall be as follows:

a. Three specimens shall receive no conditioning.
b. Three specimens shall be conditioned for 6 hours at 85° C., to determine the presence of a fugitive fungicide.

4.6.16.6 *Procedure.*—The test medium shall contain 20 gm. of agar and 1,000 ml. of distilled water. Each test specimen shall be placed in the center of a petri dish containing approximately 25 ml. of solidified test medium. The specimens shall then be sprayed with the mixed spore suspension (see 4.6.16.2). The inoculated specimens shall be incubated for 21 days at 28° to 30° C.

4.6.16.7 *Evaluation of results.*—At the end of the 21 day incubation period, each specimen shall be examined and rated as follows:

Observed fungus growth on the specimen	*Rating*
No growth	0
Traces of growth[6]	1
Partial coverage, slight to moderate growth	2
Moderate growth, considerable coverage	3
Abundant growth, complete coverage	4

[6] Traces of growth may be defined as scattered, sparse fungus growth such as might develop from an unusual mass of spores in the original inoculum, or upon an occasional extraneous bit of debris. (Continuous cobwebby growth, extending over the entire surface of the specimen, even though not necessarily obscuring the specimen, should be rated as 2.)

4.6.17 *Chemical analysis.* — A chemical analysis shall be performed to determine the relative proportions and the general type of the basic resin, the plasticizing agents, and the pigments or loading materials, comprising the finished compound. Specific gravity refractive index, where practicable and solubility, in organic soluents shall also be determined. (Benzene, petroleum ether, acetone, and carbon tetrachloride shall be considered as representing aromatic, aliphatic, oxygenated and chlorinated solvents, respectively.)

4.6.18 *Inspection procedures.*—For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** — The subject commodity shall be packaged by one or more of the following methods:

5.1.1 *Form F (film).*—Each roll shall be individually wrapped in not less than 30-pound basis weight kraft paper, waxed on one side only (waxed side out), and securely sealed with 2-inch width, water-resistant tape conforming to type III, grade B in Specification JAN–P–127. The tape shall be applied in such lengths that it will entirely surround the roll and lap on itself at least 2 inches. Rolls shall be further packaged individually in a suitable style corrugated or solid fiberboard boxes, conforming to Specification LLL–B–631 or LLL–B–636, respectively. Adequate double faced corrugated pads shall be used to protect the ends of the rolls against damage in shipment and stor-

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age. Wrapped rolls may, when required by the bureau or service concerned, be packaged individually in snug-fitting cylindrical containers, securely capped and sealed.

5.1.2 *Form T (tape).*—Tape shall be tightly wound in rolls on cores. A number of rolls placed end to end, sufficient to approximate a 24 inch length separated from each other by cardboard disks, shall be wrapped in 30-pound basis weight kraft paper, waxed on one side only, (waxed side out), and securely sealed with 2-inch width, water-resistant tape conforming to type III, grade B in Specification JAN–P–127. The tape must be applied in such lengths that it will entirely surround the roll and lap on itself at least 2 inches. Wrapped rolls of tape (approximately 24 inches in length) shall be further packaged individually in a suitable style corrugated or solid fiberboard box conforming to Specification LLL–B–631 or LLL–B–636, respectively.

5.1.3 *Form S (sheet).*—When furnished in rolls, form S insulation shall be packaged in conformance with 5.1.1 or 5.1.2, as applicable. Flat sheets when furnished shall be adequately protected against damage in shipment and storage by the use of chipboard separators. They shall be wrapped in 30-pound basis weight kraft paper, waxed on one side only (waxed side out), and securely sealed with 2-inch width, water-resistant tape conforming to type III, grade B of Specification JAN–P–127. The tape must be applied in such lengths that it will entirely surround the roll and lap on itself at least 2 inches.

5.1.4 *Form U (tubing).*—Reels of tubing shall be individually wrapped in not less than 30-pound basis weight kraft paper, waxed on one side only (waxed side out), and securely sealed with 2-inch width, water-resistant tape conforming to type III, grade B of Specification JAN–P–127. The tape shall be applied in such lengths that it will entirely surround the roll and lap on itself at least 2 inches. Tubing furnished in coils, shall be securely tied to prevent unwinding. No wrapping will be required. Reels or coils of tubing shall be further packaged in suitable style corrugated or solid fiberboard boxes, conforming to Specification LLL–B–631 or LLL–B–636.

5.2 **Packing.** — Shipping containers shall contain an identical number of items of one type, form, size and color.

5.2.1 *For domestic shipment.* — Electrical insulation, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, a wood-cleated plywood box, a nailed wood box, or a wire-bound wood box, conforming to Specification LLL–B–631, LLL–B–636, NN–B–601, NN–B–621, or NN–B–631, respectively. The gross weight of the last three types shall not exceed approximately 150 pounds.

5.2.1.1 All flaps of fiberboard boxes shall be sealed by means of a suitable adhesive, or shall be metal-stitched or taped. If flaps are taped, water-resistant tape 2½ inches wide conforming to type III, grade B of Specification JAN–P–127 shall be used.

5.2.2 *For overseas shipment.* — Electrical insulation packaged as specified in 5.1, shall be packed in wood-cleated plywood boxes, nailed wood boxes or fiberboard boxes, conforming to Specification JAN–P–105, JAN–P–106 or JAN–P–108, respectively. Plywood shall be type B, condition I, conforming to Specification JAN–P–139. The gross weight of wood boxes shall not exceed approximately 150 pounds; of fiberboard boxes, approximately 40 pounds. Each box shall be lined with a sealed waterproof bag made from material conforming to Specification JAN–P–125 for case liner material. Seams and closures shall be sealed with water-resistant adhesive conforming to Specification JAN–P–140 and shall have water resistance equal to that of the body material. Each box shall be strapped in accordance with the appendix of the applicable box specification.

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5.3 Marking.

5.3.1 *Packages.*

5.3.1.1 *Army.*—Unless otherwise specified, all Army purchases shall be marked as specified by the procuring service.

5.3.1.2 *Navy.*—Packages shall be clearly and legibly marked with the quantity, nomenclature of contents, standard Navy stock number, name of manufacturer and such other markings as may be specified in the contract or order.

5.3.1.3 *Air Force.*—Unless otherwise specified, packages shall be marked in accordance with Specification 40985.

5.3.2 *Shipping containers.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with Specification 100–2, for the Air Force in accordance with Specification 94–40645, and for the Navy in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Intended use.

6.1.1 *Type A.*—This type of material is intended for ultra high-frequency applications. It possesses excellent electrical properties which remain practically constant over a wide range of frequencies, temperatures, and humidities, and has a very low loss factor.

6.1.2 *Type B.*—This type of material is intended for applications requiring a good dielectric strength, fairly low loss factor, good moisture resistance, and good insulation resistance.

6.1.3 *Type C.*—This type of material is intended for applications requiring good dielectric strength, fairly low loss factor, and a relatively high degree of heat resistance with some sacrifice in moisture resistance.

6.1.4 *Type D.*—This type of material is intended for radio-frequency applications requiring very good dry dielectric strength, fairly low loss factor, and fair moisture resistance.

6.1.5 *Type E.*—This type of material is intended for applications having excellent mechanical strength and excellent bursting strength. It is intended for general-purpose electrical applications but is more rigid than type F.

6.1.6 *Type F, grade a.*—This type of material is intended as a high-elongation, elastic material with good electrical properties, suitable as an alternate for rubber insulating tape in cable or other applications.

6.1.7 *Type F, grade b.*—This type of material is intended as a high-elongation, elastic material similar to type F, grade a, with less rigidity at low temperatures and other characteristics required for aeronautical applications.

6.1.8 *Type F, grade c.*—This type of material is intended as a high-elongation, elastic material similar to type F, grade a, but suitable for operation at higher temperatures (105° C).

6.2 Ordering data. — Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type required. (See 1.2 and 6.7.)

c. Form required, and whether form S is to be furnished flat or in rolls. (See 1.2.)

d. Grade required for type F (see 1.2).

e. Class required (see 1.2).

f. Category required for form U (see 1.2).

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g. Color required. The following colors are generally available: black, blue, green, red, transparent (clear or slightly tinted), white, and yellow. (See 3.2.4.)

h. Dimensions required (see 3.3):

Forms F. S, and T:
Thickness and width.
Diameter of roll and mandrel.

Form U:

Number of feet required.
Number of feet in single reel or coil (see 3.3.4).

Nominal inside diameter. (See table VI).

Nominal wall thickness. (See table VI).

Size number for AWG-size tubing (see table VI).

i. Whether form T is for application by hand or machine. (See 3.3.3.)

j. Number of pounds required for forms F, S, and T.

k. Whether the insulation is to be packed and marked for domestic or overseas shipment. (See 5.2 to 5.3.2, inclusive.)

l. For Army purchases, marking of packages. (See 5.3.1.1.)

6.2.1 Requests, requisitions, and schedules should specify that bidders shall submit with their bids a statement in triplicate showing:

a. Diameter of roll or reel in inches.

b. Weight of roll or reel.

c. Number of feet per roll or reel.

d. Number of feet per pound of material.

6.3 In the procurement of products requiring qualification the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

6.4 Fungus resistance.—The requirements and tests for fungus resistance are designed to select funginert materials, that is, materials that are permanently resistant to fungus growth because they do not furnish nutriment to these organisms. For this reason, the tests (see 4.6.16) are intended both to determine whether fungi can grow on the materials and whether lack of such growth may be due to fugitive fungistatic materials in the specimens.

6.5 Sources of stock fungus cultures.—The test fungi designated for use in tests may be obtained from the American Type Culture Collection, 2029 M Street, N. W., Washington 6, D. C.

6.6 Signal Corps restriction on use of "spaghetti" tubing.—Form U insulation, often called "spaghetti" tubing, shall be used on Signal Corps equipment only with the express approval of the contracting officer.

6.7 Applications for type F, grade a tape.—For the information of purchasing activities, applications for type F, grade a tape are at the present time limited to thickness of 0.010 inch and larger. The approximate number of feet per pound of type F, grade a tape is as specified in table XII.

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TABLE XII.—*Conversion table for tape*

Thickness	Width	Length of tape per pound (approximate)
Inch	*Inches*	*Feet*
0.010	0.50	360
.020	.50	180
.010	.75	240
.020	.75	120
.010	1.00	180
.020	1.00	90
.010	1.25	144
.010	1.50	120

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy—Bureau of Ships

Other interest:
Army—OS
Navy—AOS
Air Force.

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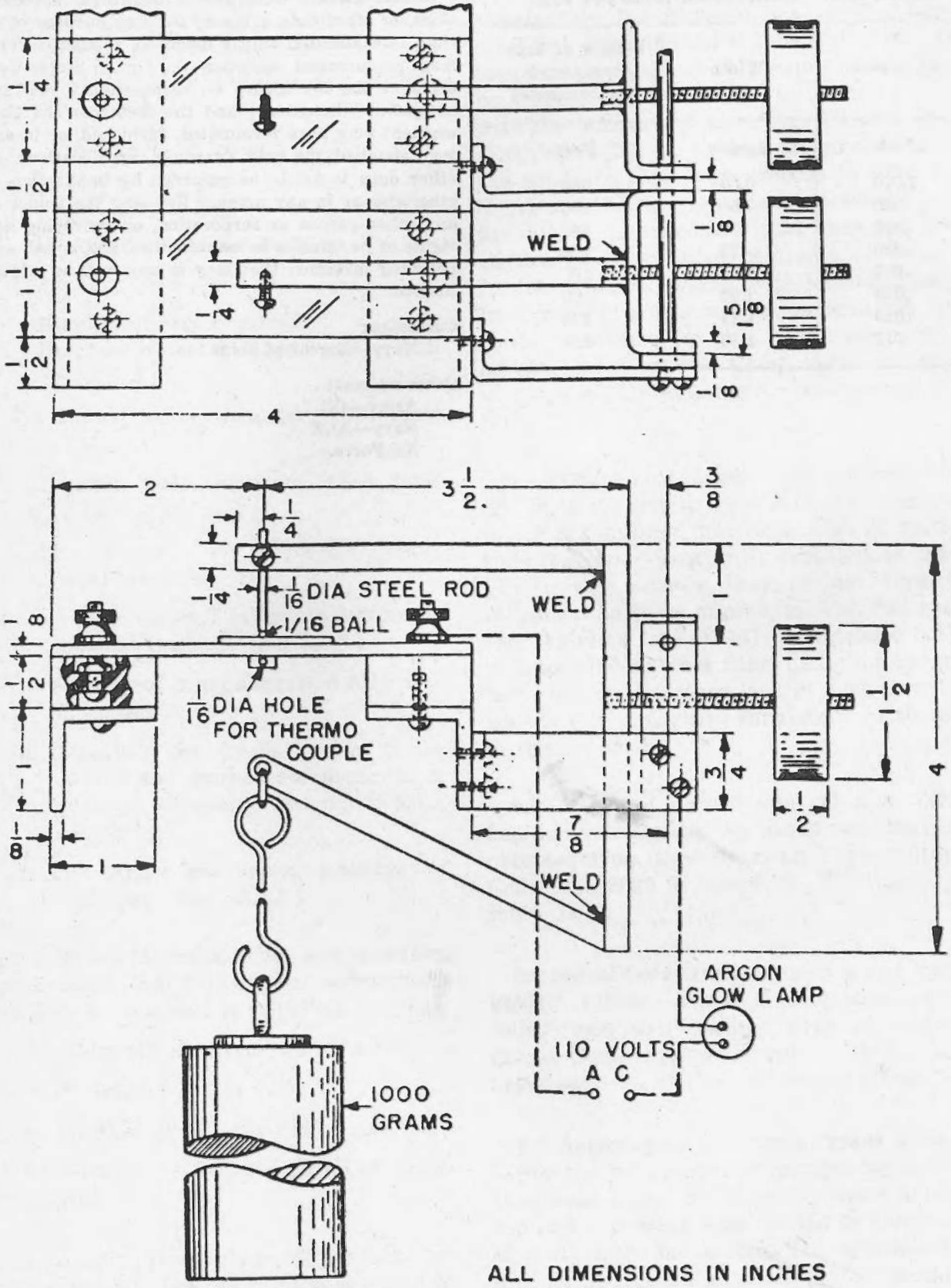


FIGURE 1.—*Equipment for softening—temperature (penetration) test.*

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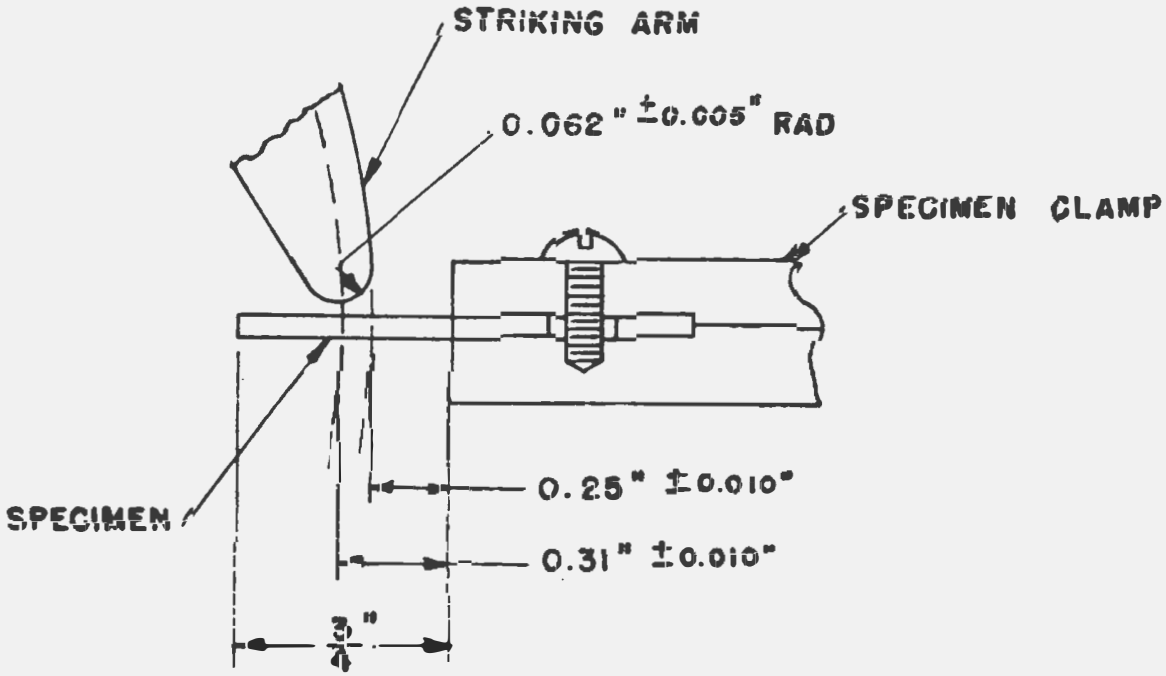


FIGURE 2.—*Specimen clamp and striking arm.*

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